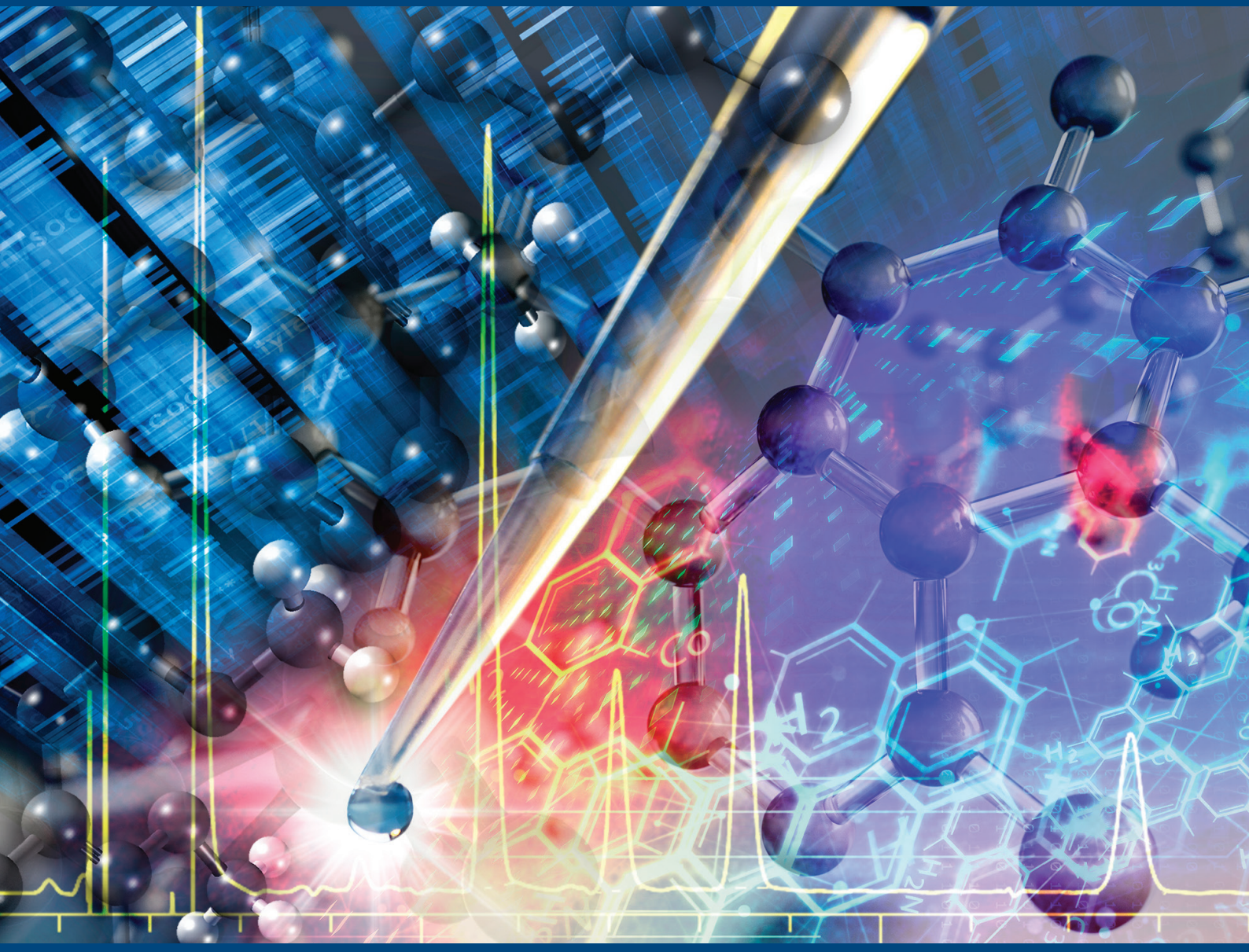
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Pharmaceuticals, Pesticides, and PFAS: Quantifying Endocrine Disrupting Compounds in Plastics and Fish Tissues Using Solvent Extraction and LC-MS/MS

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ABSTRACT

The rise of plastic pollution in marine environments has been heavily documented, with particular focus on the physical impacts the plastics can have on biota. But, plastics also sorb a range of hydrophobic chemical pollutants, acting as vectors for the transportation of these compounds throughout marine environments. Therefore, an analytical method that can target both marine biota and plastic matrices will be key to advance our understanding of the link between chemicals in the environment, plastic pollution, and effects on biota. Here, an efficient method for the detection and quantification of a broad suite of compounds in marine samples was developed. Five extraction methods were trialed for the analysis of 21 pesticides, PFAS, and pharmaceuticals in biota and plastics. This included three ultrasonic extraction methods and two QuEChERS methods. Ultrasonic extraction in acetonitrile with a microcentrifuge step then concentration by Bond Elut Carbon SPE resulted in best recovery across most compounds. Of the 21 compounds trialed, 16 were efficiently quantified. Method limits of quantification and detection were between 0.02 and 4.81 ppb (mLODs) and between 0.06 and 14.60 ppb (mLOQs). This method is widely applicable to a range of marine environments and supports routine evaluations of environmental safety and monitoring protocols.

1 | Introduction

Plastic pollution has become ubiquitous in the marine environment [1]. While research on the occurrence of plastics in aquatic environments is thriving [2, 3], our understanding of its chemical impacts and risk of harm is still limited [4]. Plastics can sorb hydrophobic contaminants, including persistent organic pollutants (POPs) at concentrations up to 500 times higher than the underlying water column [5, 6]. Once contaminants are sorbed into plastics, they can be transported by currents away from their source sites and potentially consumed by marine organisms. Therefore, plastics are a vehicle to deliver a cocktail of chemical contaminants to biota [7], with plastic in gastrointestinal tracts linked to increased concentrations of endocrine disrupting compounds (EDC) [8, 9]. This association between plastics and chemical contaminants could amplify bioaccumulation and magnification effects in food webs, compounded by the potential translocation of micro- and nanoplastics across physiological membranes [10].

Chemical contaminants such as pharmaceuticals, personal care products, pesticides, and per- and polyfluoroalkyl substances (PFAS) are ubiquitous across coastal, marine, and terrestrial environments. This is due to their continued and increased use in human and veterinary medicine, agriculture, and industry [11, 12]. Many of these compounds are classed as POPs as they are

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hydrophobic, lipophilic, and do not degrade in the environment [13]. In addition, they are classed as EDCs because they negatively impact the endocrine system by mimicking hormones or blocking secretion, transport, binding, or natural hormone production [14, 15]. Organisms exposed to EDCs experience changes to natural homeostasis, reproduction, and behavior; therefore, there is a growing concern about the impact of POPs and EDCs on marine organisms [16–18].

The presence of pharmaceutical compounds and pesticides, such as fluoxetine, atrazine or diclofenac, in aquatic environments decreases fish growth and elicits a suite of biochemical responses [19–21]. The surfactant-like properties of PFAS drive its accumulation in the liver of marine organisms [22–24]. There are calls to prioritize our understanding of contamination levels, ecotoxicity, and transfer along the food web, as PFAS triggers altered organ function and growth, among many other responses (e.g., reduced water filtration, oxidative stress, or alterations to gonad and digestive glands) [25, 26]. Overall, concerns about the persistence of PFAS in the environment and its suite of negative impacts to biota and humans have led the US government to drastically decrease health advisory limits for PFAS [27]. Many jurisdictions are also banning, limiting, or attempting to mitigate known sources of PFAS and other EDCs.

EDCs are released from multiple sources into the marine environment. In addition to industrial sources, pesticides can be washed into waterways by runoff from farmland or household yards, as pharmaceuticals and personal care products are often not fully metabolized by the body and, therefore, released through human and animal excretion [16, 17, 28]. PFAS are commonly used in firefighting foams, textile additives, and food packaging due to their stain and water repellent properties [29, 30]. In tandem, a major concern and uptake route for biota is the enhanced association of many EDCs with plastics because of their ubiquitous presence in aquatic environments [31, 32].

Due to the rise in plastic and chemical pollutant occurrence, an analytical approach that can target and compare biota and plastic samples will facilitate a better understanding of the link between chemicals in the environment, plastic pollution, and associated effects on biota. While many studies look at individual compounds (compound-specific approaches), we lack an efficient multiresidue analytical method for the simultaneous assessment of multiple pesticide, pharmaceutical, and PFAS in a range of complex matrices such as environmental samples [33]. Such a method would provide a versatile and potentially cost-effective tool to support broad and long-term monitoring of chemical compounds. Liquid chromatography mass spectrometry (LC-MS) with a preliminary extraction step to concentrate compounds in environmental matrices is a common approach for extraction of a range of compound classes [34–37].

Many extraction techniques have been used to effectively analyze compounds from the same class, including QuEChERS (quick, easy, cheap, effective, rugged, safe) extraction and cleanup [35, 38, 39], ultrasonic extraction [40, 41], accelerated solvent extraction [22], Soxhlet extraction [42], pressurized solvent extraction [43], liquid–liquid extraction [38], or microwave-assisted extraction [44]. Optimization steps generally include variations in, for example, solvent, run time, and number of extractions, to provide the most efficient extraction for different compounds and matrices [35, 36]. Attention to matrix effects and removing interferences is also important in tissue samples due to the coextraction of lipids and proteins along with the analytes of interest. Overall, while multiple approaches have been used, ultrasonic and QuEChERS extractions are prevalent and widely used across a suite of environmental samples [35, 45, 46]. Both approaches have demonstrated applicability to complex matrices, including tissues, and offer flexibility to optimize the extraction method and minimizing matrix effects through changes in solvents and pre- and postextraction steps.

The present study aims to develop and validate a multiresidue analytical methodology for the detection and quantification of a suite of pharmaceuticals, pesticides, and PFAS in plastic and fish tissues. Developing methods to analyze a suite of compounds in tissue as well as plastics is critical to further investigating the transfer of chemical pollutants from ingested plastics. A range of extraction methods were trialed and optimized for the two matrices to enhance compound recovery. Specifically, the performance of an ultrasonic extraction method across two solvents was evaluated, as well as the implementation of several cleanup steps including a microcentrifuge step and two solid phase extraction (SPE) cartridge types. In addition, the QuEChERS method, marketed as a quick, easy, cheap, and rigorous method for extraction [47], was tested. While the QuEChERS method has been widely used for the extraction of pesticides and pharmaceuticals from complex environmental samples, including marine biota [35, 48, 49], it has not been evaluated for extraction from microplastics. Recovery performance was validated using liquid chromatography tandem mass spectrometry (LC-MS/MS).

2 | Materials and Methods

2.1 | Target Contaminants

A group of 21 different compounds, including pharmaceuticals, pesticides, and PFAS, was selected based on a range of criteria, including but not limited to their widespread use in Australia and their level of regulatory concern [50]. The 21 compounds of interest were pharmaceuticals including amoxicillin, ibuprofen, estrone (E1), 17β-estradiol (E2), 17α ethynylestradiol (EE2), atenolol, hydrochlorothiazide, tramadol, carbamazepine, and diclofenac; pesticides including atrazine, metabolites atrazine-desisopropyl, atrazine-desethyl, simazine, 2,4-D, MCPA, diuron, and tebuconazole; and PFAS including perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and perfluorohexanesulfonic acid (PFHxS; Table S1 summarizes logK_{ow} values, compound structure, formula, and CAS number). All selected reaction monitoring (SRM) information can be found in Table S2.

2.2 | Reagents and Materials

Tested compounds tebuconazole; 2,4-dichlorophenoxyacetic acid (2,4-D); atrazine; atrazine-desethyl; atrazine-desisopropyl; 3-(3,4-dichlorophenyl)-1,1-dimethylurea (Diuron); 2-methyl-4-chlorophenoxyacetic acid (MCPA); simazine; amoxicillin; carbamazepine; atenolol; hydrochlorothiazide; E1; E2; EE2; diclofenac; ibuprofen; tramadol; PFOS; PFOA; and PFHxS

were purchased from Sigma-Aldrich (St. Louis, MO, USA; CAS numbers and empirical formula are listed in Table S1). Native multicomponent solutions were prepared in methanol and kept at −20°C. A 100 ppb spiking solution was prepared for spike recovery estimates. A 100 µL aliquot of spiking solution was added to samples, resulting in a final concentration of 10 ppb for analysis. Magnesium sulfate, PSA (primary/secondary amine) bonded silica, and discovery DSC-18 were also purchased from Sigma-Aldrich (St. Louis, MO, USA). High purity, LC-MS grade methanol and acetonitrile (ACN) were purchased from Thermo Fisher Scientific (Waltham, MO, USA). Glacial acetic acid and ammonia were purchased from Sigma-Aldrich (St. Louis, MO, USA). QuEChERS (7 g extract pouch with ceramic homogenizer) kits and Bond Elut Carbon SPE cartridges (100 mg, 1 mL) were purchased from Agilent Technologies (Santa Clara, CA, USA). Oasis HLB (hydrophilic and lipophilic balance) SPE cartridges (3 cc) were purchased from Waters Corp. (Milford, MO, USA). Ultrapure water was obtained from a Milli-Q reagent water system (Merck, Darmstadt, Germany).

2.3 | Sample Collection and Preparation

Two fish species, Australian herring (*Arripis georgianus*) and salmon trout (*Arripis trutta*), were collected from fish markets across South Australia from 2018 to 2020. Fish were transported on ice and in the dark to the laboratory, where dorsal muscle tissue as well as livers were dissected. Samples were frozen at −20°C then freeze-dried using a Labconco 4.5L Benchtop Freeze Dryer, homogenized using a ball mill and stored at −20°C for subsequent extraction and analysis.

Polypropylene plastic samples were obtained from Plasdene (South Australia, Australia) and shredded to < 3 mm (microplastics) using an industrial shredder and 3 mm sieve. Microplastics were then placed in nitex nylon bags and submerged in the marine environment for 12 months. This process was conducted to ensure plastic samples used in method development accurately reflect microplastics found in the environment under conditions such as weathering, degradation, and biofouling. This will be critical for method uptake investigations into the impacts of microplastics in the environment. Upon collection, plastics were triple-rinsed with ultrapure (Milli-Q) water to remove any biological material and dried in a laminar flow cabinet. Due to the hydrophobic nature of the chemical contaminants being analyzed, this is unlikely to impact their association with the plastic surface. Replicate microplastic samples of 0.5 g were stored at −20°C for subsequent extraction and analysis.

2.4 | Method Trials

Five different trials were run to identify and optimize the best extraction process for the selected compounds, tissues, and plastic samples (Figure 1).

2.4.1 | Sample Pretreatment

A 0.5 g portion of freeze-dried and homogenized muscle tissue, a 0.1 g portion of freeze-dried and homogenized liver, or a 0.5 g portion of dried microplastics were weighed into separate 15 mL polypropylene tubes for Trials 1–3. For Trials 4 and 5, a 0.1 g portion of freeze-dried and homogenized liver or muscle tissue or a 0.5 g portion of dried microplastics were weighed into separate 50 mL polypropylene tubes (in Trial 5, the tubes provided in the kit were used). Each sample was spiked with the multicomponent spiking solution outlined above, giving a final concentration of 10 ppb for analysis. The fish tissues used in the method development component of this study were collected from fish markets and were caught in offshore locations. Due to the nature of the environment in which they were caught, the likely baseline contamination levels are extremely low and will therefore make up an extremely low proportion of the signal being seen for the 10 ppb spiked multicomponent solution. Each trial had between 6 and 15 replicates for each matrix.

2.4.2 | Trial 1—Ultrasonic Extraction With MeOH and Bond Elut SPE

A 4 mL aliquot of methanol was added to the samples and ultrasonicated in an ultrasonic bath at 25°C for 20 min. Following ultrasonication, the samples were centrifuged at 2500 × *g* for 15 min. The supernatant was transferred to a clean 15 mL polypropylene tube, and the extraction procedure was repeated another two times with methanol containing 1% ammonia. A total of 12 mL of supernatant was collected across the three extraction cycles. The supernatant was then concentrated to 1 mL under a gentle stream of nitrogen (N_2), then 200 µL glacial acetic acid was added and vortexed for 30 s. Supernatants were reconcentrated using SPE. Supernatants passed through Bond Elut Carbon SPE using a polypropylene syringe. The cartridge was then rinsed with methanol and the extract was concentrated to 1 mL with a gentle stream of nitrogen for LC-MS/MS analysis.

2.4.3 | Trial 2—Ultrasonic Extraction With ACN and Bond Elut SPE

The ultrasonication and centrifugation were performed as in Trial 1, with ACN containing 1% ammonia as the extracting solvents. After this, the 12 mL of supernatant was evaporated to dryness and reconstituted in 1 mL of methanol. The solution was transferred to a 2 mL Eppendorf tube and centrifuged at 12 000 rpm for 5 min. The supernatant was then decanted again into a 2 mL Eppendorf tube where 200 µL of glacial acetic acid was added and the tube vortexed for 30 s. The tube was then vortexed for 15 min at 4000 rpm. Supernatants were then passed through a Bond Elut Carbon SPE cartridge using a polypropylene syringe. The SPE cartridge was rinsed with an additional 700 µL of methanol, and the extract was concentrated to 1 mL under a gentle stream of nitrogen for LC-MS/MS analysis.

2.4.4 | Trial 3—Ultrasonic Extraction With ACN Solvent and HLB SPE

Ultrasonic extraction was completed as per Trial 2 above, though variation in the SPE step occurred. The sample was diluted to 10% in methanol, then passed through a HLB SPE cartridge.

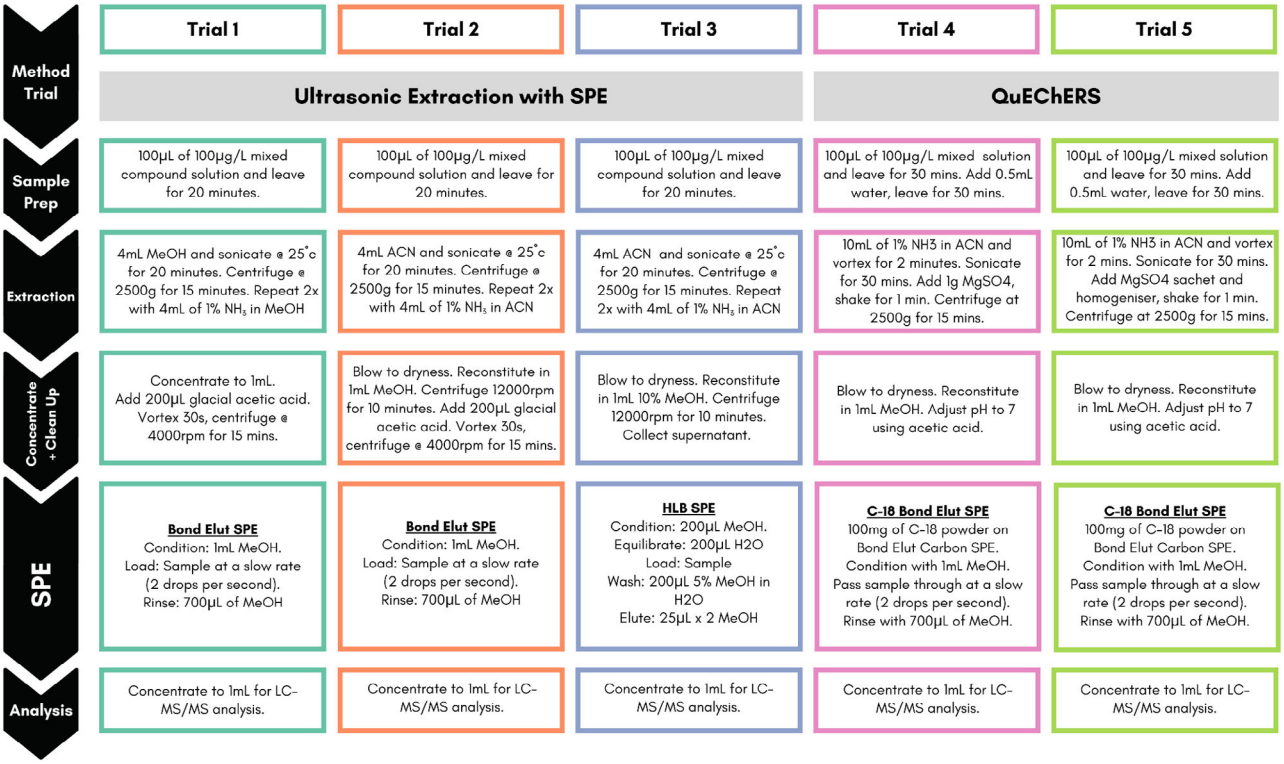


FIGURE 1 | A summary of the methods trialed throughout the study, showing key differences in critical steps.

The cartridge was first conditioned with 1 mL of methanol and equilibrated with 1 mL of H_2O. The spiked sample was then loaded onto the HLB cartridge, washed with 5% methanol in water, and eluted using two aliquots of 25 µL of methanol. The elution was then concentrated to 1 mL under a gentle stream of nitrogen for LC-MS/MS analysis.

2.4.5 | Trial 4—QuEChERS Extraction Manual

Both QuEChERS protocols were based on Anastassiades et al. [51] with modifications. The sample was wet with 0.5 mL of ultrapure (Milli-Q) water and left to equilibrate for 30 min. A 10 mL aliquot of 1% ammonia in ACN was added and vortexed for 2 min. After ultrasonication (ultrasonic bath, 25°C for 30 min), 1 g of magnesium sulfate was added, and the mixture was shaken vigorously for 1 min. The mixture was centrifuged at 2500 × *g* for 15 min to allow separation of the organic layer from the sample. The organic layer was then decanted into a 15 mL centrifuge tube and concentrated to 1 mL under a gentle stream of nitrogen. The pH of the solution was then checked and adjusted to 7 with glacial acetic acid. The solution was then passed through Bond Elut Carbon SPE that had been preprepared with a 100 mg layer of C18 powder on top. The cartridge was then rinsed with methanol and the extract was concentrated to 1 mL with a gentle stream of nitrogen for LC-MS/MS analysis.

2.4.6 | Trial 5—QuEChERS Extraction With QuEChERS Kit

Wetting and ultrasonication of the sample were performed as outlined in Trial 4. Following ultrasonication, the sachet of magnesium sulfate (7.5 g, 6 g magnesium sulfate and 1.5 g sodium chloride; Agilent Technologies, CA, USA) was added as well as the ceramic homogenizer and the tube was shaken vigorously for 1 min. The homogenizer was removed, and the mixture was centrifuged at 2500 × *g* for 15 min to cause separation of the organic layer from the sample. The organic layer was decanted into a 15 mL centrifuge tube and then concentrated to 1 mL under a gentle stream of N_2. The pH of the solution was then adjusted to 7 by the addition of acetic acid. The solution was then passed through Bond Elut Carbon SPE that had been preprepared with a 100 mg layer of C18 powder on top. The cartridge was then rinsed with methanol and the extract was concentrated to 1 mL with a gentle stream of nitrogen for LC-MS/MS analysis.

2.5 | Environmental Matrices Analysis

To verify the methods in environmentally relevant samples, we collected the following three species of fish from Port River (South Australia): yellow-eyed mullet (*Aldrichetta forsteri*), Australian herring (*A. georgianus*), and King George whiting (*Sillaginodes punctatus*) during 2022. In addition, we also tested the methods on polypropylene plastic samples that had been submerged at same sites as fish locations across the Adelaide metro coastline (South Australia) for a period of 70 days during 2022.

2.6 | Instrumental Analysis

Analyses were performed by LC-MS/MS using a Dionex UltiMate 3000 UHPLC+ (Thermo Fisher Scientific, MO, USA) connected to a TSQ Altis triple quadrupole with an electrospray source interface (ESI; Thermo Fisher Scientific, MO, USA).

Data were processed using Thermo TraceFinder Software for quantitative and qualitative analysis (Thermo Fisher Scientific, MO, USA). The chromatographic column was a Kinetex C18 (100 × 2.1 mm × 2.6 μm particle size; Phenomenex, CA, USA). The mobile phases consisted of (A) methanol and (B) water. The following gradients were applied: 0 min (2% A), 3.5 min (2% A), 5 min, (95% A), 7 min (95% A), 9 min (95% A), and 10–15 min (2% A) and returned to initial conditions. The flow rate was kept at 0.25 mL/min for the first two time segments, then 0.35 mL/min thereafter. The sample volume injected was 10 μL, and analysis was carried out in both positive and negative ion modes. Data acquisition was carried out in SRM to both identify and quantify analytes using two precursor–product ion transitions, retention times, and the ratio of the intensities of the two product ions [52, 53]. Collision energies were optimized for each analyte (Table S2).

2.7 | Quantification and Quality Control

Analytical performances and quality control were evaluated for each run by solvent blanks (MeOH) and procedural blanks to evaluate contamination and detection limits in accordance with US EPA Method 1633 and amendments [54]. Linearity was evaluated for each analyte using calibration curves (range between 0.05 and 500 ppb; Table S3). Quantification was carried out using parent and product ion transition with the use of the quantitative/confirmation ion ratio (Table S2; Figure S1a–c). The precision of the trial was evaluated by calculating standard error across replicates in each trial.

The performance of each trial was determined by evaluating the recovery of spiked concentrations from each matrix. Spike relative recovery was calculated using the following formula:

$$\text{Relative recovery} = \frac{C\,(\text{spiked})}{C\,(\text{matrix})} \times 100,$$

where C(spiked) is the concentration in the extracted spiked matrix and C(matrix) is the concentration in the spiked matrix.

To evaluate matrix effects each of the target compounds were spiked into the matrices (liver, tissue, and plastic) at 10 ppb and then subjected to the five procedures being trialed. Matrix effects were calculated using the following equation:

$$\text{ME}x(\%) = \left(\frac{Ax\,(\text{matrix}) - Ax\,(\text{blank})}{Ax\,(\text{solvent})} - 1\right) \times 100,$$

where Ax(matrix) is the area of compound x in the spiked matrix, Ax(blank) is the area of compound x in the nonspiked matrix, and Ax(solvent) is the area of compound x in the spiked procedural blank.

Instrument limit of quantification (iLOQ) and limit of detection (iLOD) were calculated using the following formula [55]:

$$\text{iLOD} = \frac{3.3\sigma}{S} \text{ and iLOQ} = \frac{10\sigma}{S}$$

where σ is the standard deviation of the response of multiple injections at the lowest linear concentration of the calibration curve (degrees of freedom = 7) and S is the slope of the calibration curve. Method LODs and LOQs (mLODs and mLOQs) were calculated by relating the iLOD to the recovery percentage of spiked sample.

3 | Results and Discussion

Five methods were trialed across the three complex environmental matrices with varying results. Extraction was generally much more successful across plastic trials than the two tissue matrices, with several compounds across the trials showing adequate recoveries (40%–120%; Figure 2). Enhanced results for plastics were due to the decreased complexity of the matrix and fewer coextracts being present at retention times like analytes. Recoveries across QuEChERS trials were low (40% or less) across most compounds in both muscle and liver trials, while the ultrasonic extraction methods were acceptable (40%–120% recovery) across five or more compounds for both muscle and liver tissues. Trial 2 was particularly successful for plastics, with 10 compounds recovered in the 40%–120% range.

3.1 | LC-MS/MS Optimization and Validation

Two different mobile phases were tested for use in the LC-MS/MS method, methanol and ACN. A methanol mobile phase was found to be more suited to the compounds of interest due to increased linearity of calibration curves (Table S3).

Quality control and quality assurance measures were taken throughout LC-MS analysis to ensure consistency and validity of results. Methanol and method blanks were analyzed throughout and showed no signs of contamination in the LC-MS system or throughout the method trials.

Linearity was evaluated using a 9-point calibration curve, and linearity constants of 0.97 (atrazine-desisopropyl)–0.99 were achieved for all compounds investigated except amoxicillin, E1, E2, EE2, and hydrochlorothiazide (Table S3). These compounds were not well-suited to the developed LC-MS method and would likely benefit from the use of an alternate column and mobile phase. Previous studies have used mobile phases such as ammonium acetate in ACN or aqueous ACN for the analysis of antibiotic and estrogenic compounds with favorable results [56, 57]. While the importance of estrogenic and antibiotic analysis in environmental samples is recognized, the optimization of a broad range of compounds across multiple matrices was prioritized over optimizing for individual compounds within this method. The optimization of the amoxicillin and estrogenic analysis within this LC-MS method was not investigated further, though there is a potential for their inclusion in a multistage analytical method in future studies. iLOQs and iLODs were found to be between 0.034–0.188 ppb (iLOD) and 0.102–2.443 ppb (iLOQ; Table S2).

The US EPA standard for acceptable PFAS recovery, as well as other examples in the literature, were used as a basis for the parameters of acceptable recovery [36, 40, 54, 58, 59]. Trials were considered acceptable if an average recovery between 40% and 160% was achieved as per similar trends seen in literature [36, 40, 54, 59]. Results below 40% recovery were considered unfavorable whereas results over 200% were considered unacceptable and

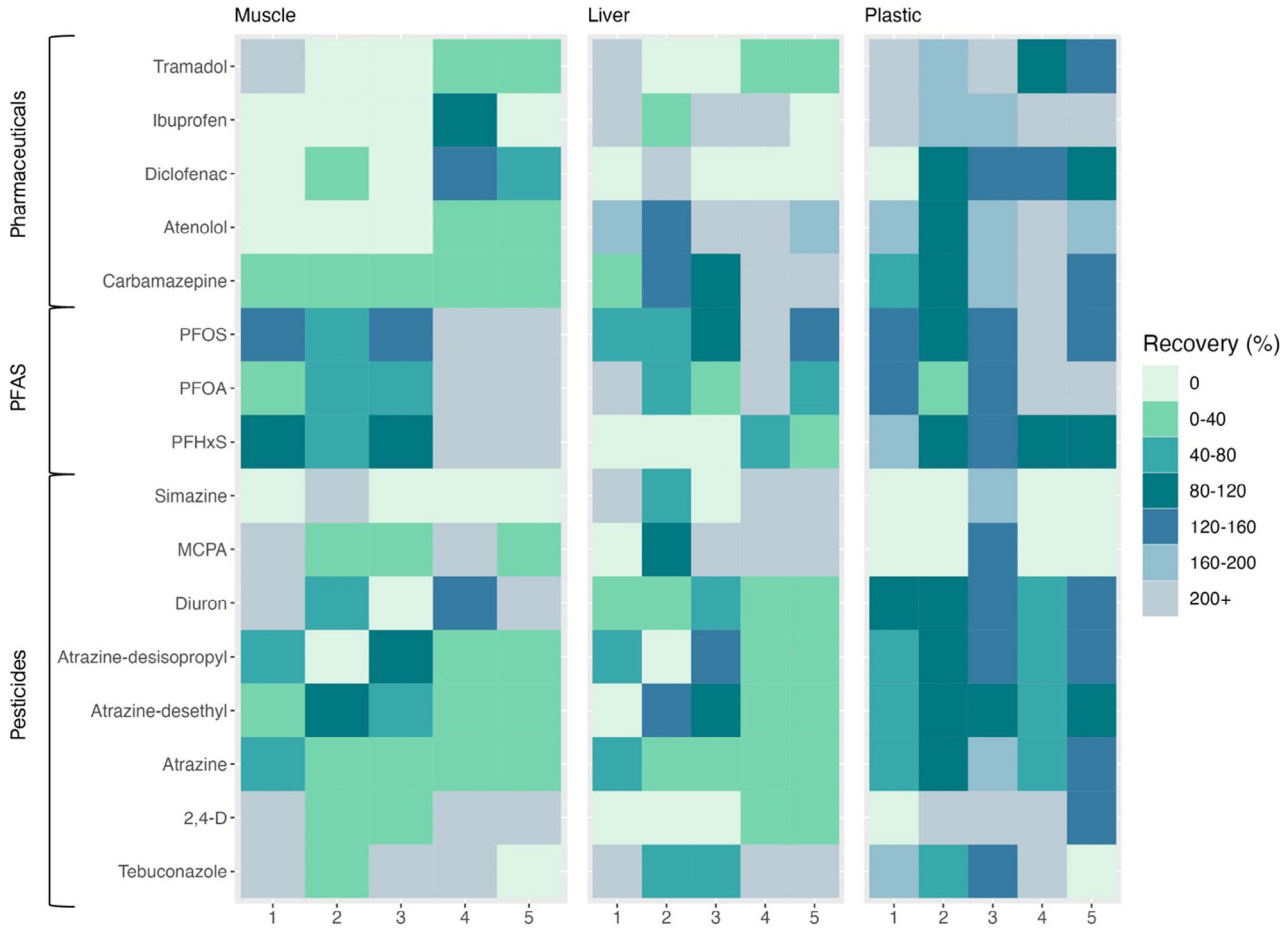


FIGURE 2 | Relative recovery performance of each method trial (1–5) for each of the compounds tested. Relative recovery is presented as a percentage and compounds are ordered by their use class.

removed from representation in Figure 3. Results over this threshold contain a high degree of variability likely due to matrix interferences and are therefore unreliable for quantification.

3.2 | Extraction Method Trials Selection and Optimization

As the method trials were conducted across a range of environmental matrices and a range of compounds of interest, the results were varied as can be seen in Figure 3.

3.2.1 | Trial 1—Ultrasonic Extraction With MeOH and Bond Elut SPE

This trial was effective for extraction of PFAS from muscle tissue, where the relative recovery of PFOS and PFHxS were 139.8% and 83.6%, respectively (Figure 3; Table S4), falling within the acceptable recovery bounds of US EPA method 1633 [54]. There was reasonable recovery for atrazine and its metabolites from muscle tissue, with recoveries of 65.6% and 69.7% for atrazine and atrazine-desisopropyl, respectively. This trend was also seen in extraction from liver tissue during Trial 1, with the most successful recoveries being PFOS, PFHxS, atrazine, and atrazine-desisopropyl (Figure 3). Across both liver and muscle tissue, there was no recovery of atenolol, diclofenac, ibuprofen, and simazine, and significant signal enhancement was seen for MCPA, diuron, tebuconazole, and 2,4-D (Figure 3). During the extraction process in Trial 1, visible coextracts were present in both liver and muscle samples. These were most likely lipids and proteins and while care was taken to remove these throughout, coextracts were present and likely interfered with the analysis [60]. Therefore, it was identified that an additional cleanup step was needed for both liver and muscle tissues to remove these coextracts, as well as the evaluation of an alternate solvent. In extraction from plastics, similar trends were seen in the tissue samples, where PFAS and atrazine metabolites were within successful limits. Ibuprofen, carbamazepine, MCPA, and 2,4-D were also recovered successfully from plastic samples, in contrast to the results for tissue samples under the same extraction conditions.

3.2.2 | Trial 2—Ultrasonic Extraction With ACN and Bond Elut SPE

For Trial 2, the use of ACN and the microcentrifuge step were aimed at eliminating coextracts seen in Trial 1. There was visible precipitation of lipids/proteins from both liver and muscle tissues in this step, resulting in a cleaner sample for analysis due to the use of ACN and the microcentrifuge step. Again PFOS, PFOA, and PFHxS were all extracted successfully from muscle and liver

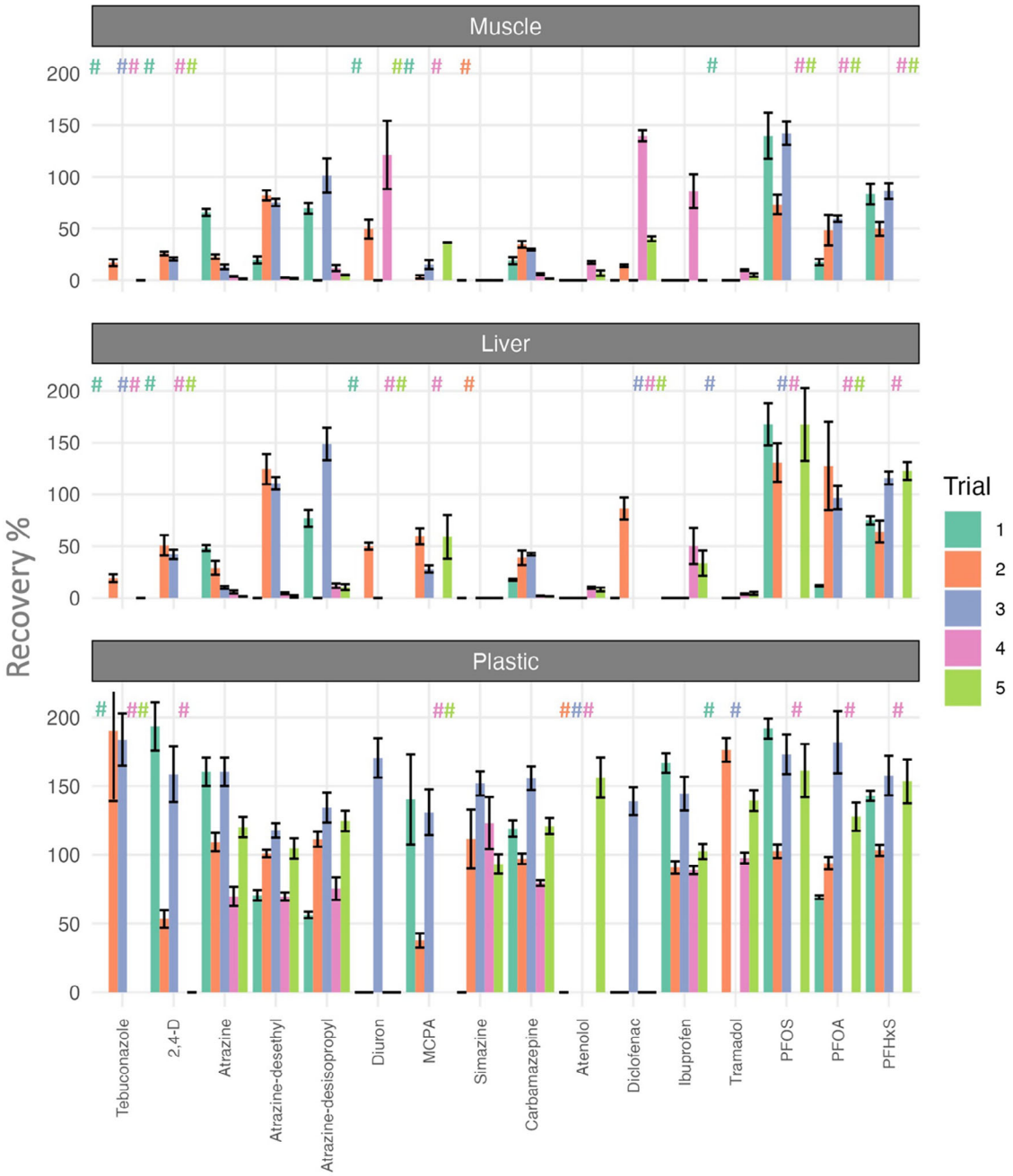


FIGURE 3 | Influence of extraction procedure (Trials 1–5) on relative recovery at spiking level 10 ng/g ($n = 6$–15). Only results below 200% recovery are shown and compounds amoxicillin, E1, E2, EE2 have been removed due to poor performance. # indicates trials where average recovery was above 200%. See Table S4 for full details.

tissues (Figure 3 Table S4). In addition, diuron and atrazine-desethyl recoveries were both more successful than Trial 1 for both tissue types (from results outside of successful limits to 49.7% and 82.2% for muscle and 50.0% and 124.3% for liver, respectively). In liver tissues, there was also an increase in the recoveries of 2,4-D, MCPA, and diclofenac. Conversely, this method saw significant signal enhancement of simazine in both tissue types to 650.84% and 791.88% in muscle and liver tissue, respectively.

Lipids are often coextracted with contaminants due to their high solubility in polar solvents such as methanol, though solubility in ACN is lower [61, 62]. Removing lipids from the extract without removing the lipophilic analytes is key to ensuring the highest recovery of analytes possible from the complex matrix [62]. The increase in recovery for several compounds (e.g., atrazine-desethyl, diuron, carbamazepine, and PFOA) in the muscle and liver matrix during Trial 2 may be attributed to the increased efficiency in the removal of lipids by changing the extraction solvent to ACN and the addition of the microcentrifuge step to further elute lipids and proteins.

There was successful extraction from plastics across a range of compounds including all PFAS, tramadol, ibuprofen, carbamazepine, simazine, atrazine and its metabolites, 2,4-D, and tebuconazole. As the plastic does not contain lipids and proteins like the tissues, there is no benefit from the microcentrifuge step;

therefore, it is likely that the change of extraction solvent (MeOH to ACN) was responsible for the favorable recoveries from plastics in this trial. Diuron and MCPA had low recoveries in this trial, 0% and 37.8%, respectively.

3.2.3 | Trial 3—Ultrasonic Extraction With ACN Solvent and HLB SPE

As the name suggests, HLB SPE contains a hydrophilic- and lipophilic-balanced copolymer and is designed to reduce interferences for the extraction of a wide range of polar analytes [63]. While theoretically more advantageous for the extraction of polar compounds compared to Bond Elut SPE, we did not see improved efficiency. In muscle tissues, there was favorable recovery of PFAS and atrazine metabolites as seen in previous trials but no pharmaceuticals were recovered, except for carbamazepine. Of the remaining pesticides, 2,4-D and MCPA recovered at low percentages (20.7% and 15.3%, respectively). Similarly, in liver tissues, there was adequate recovery of PFAS, except for PFOS where significant signal enhancement occurred (324.5% recovery). There was an increased recovery of pesticides 2,4-D and MCPA in comparison to the muscle tissues, though recoveries were still lower than Trial 2 (42.02% and 28.00%, respectively). Atrazine metabolites were also recovered with minimal signal enhancement (110.7% and 148.8%, respectively). HLB SPE extraction from a plastic matrix was much more successful. All compounds were extracted with high efficiency, except for atenolol and tramadol, which showed strong signal enhancement (1181.6% and 2564.1% recoveries, respectively).

Overall, the complexity of the tissue matrix and potential interferences from lipid and protein coextracts still limited the extraction and recovery of several compounds in muscle and liver tissues. On the other hand, the efficacy of HLB SPE was noticeable in the less complex matrix such as plastics. Previous studies have used HLB SPE for the extraction of a range of pharmaceutical compounds from various tissues in fish with high variability in recoveries ranging from 46% to 113% and 68% to 107% [61, 64]. This variability is seen across several studies and can be attributed to the complexity of the matrix.

3.2.4 | Trial 4—QuEChERS Extraction Manual

QuEChERS extraction is widely and effectively used for the extraction of pesticides and other residues from food samples, that is, samples with complex matrices [65]. The results indicate that while the manual QuEChERS approach was advantageous for diuron, diclofenac, and ibuprofen, other compounds either exhibited extreme signal suppression or enhancement, indicating interference from coextracts. This was seen especially in PFAS where signal enhancement was close to 3000% for PFHxS. A similar result was seen in liver tissues where significant signal enhancement was present across PFAS. Ibuprofen was the only compound to be extracted within the successful limits from the liver tissue (50.19%) whereas all other compounds were either enhanced well beyond limits or extracted inefficiently below limits. In the less complex plastic samples, recovery of pesticides was successful. In particular, atrazine and its metabolites and simazine were extracted at levels close to 100% recovery. Other pesticides, 2,4-D and MCPA, showed some signal enhancement, with average recoveries approximately three times the spiked levels. PFAS were again at much higher levels than expected, showing roughly three to four times the spiked concentration.

Trial 4 with manual extraction and QuEChERS was advantageous for pesticide compounds in plastic but had limited success in the more complex matrices of fish tissues. This could be due to the magnesium sulfate:salt ratio used in the QuEChERS method for the removal of water. In this trial, magnesium sulfate was used without the addition of salt. Different magnesium sulfate:salt ratios have been advantageous for the extraction of pesticides in aquaculture products such as fish, crabs, and crayfish [45]. Extraction efficiency was found to increase across a range of pesticides with the addition of salt, as a number of these are hydrophilic and the presence of water in the matrix may result in lower extraction efficiency [45]. This was the rationale for running the subsequent Trial 5.

3.2.5 | Trial 5—QuEChERS Extraction With QuEChERS Kit

The QuEChERS kit includes the magnesium sulfate cleanup sorbent preweighed and mixed with sodium chloride in order to control polarity and has been shown to increase the efficiency of extraction of key analytes [45, 66]. Here, the QuEChERS kits performed similar to the manual QuEChERS extraction. In muscle tissues, no compounds had recovery within ±25% of the spiked concentration, and signal enhancement was seen across PFAS as well as 2,4-D and diuron (approximately 290%–600%). In liver tissues, there was an increased recovery of MCPA and two of the three PFAS were within a reasonable range of the spiked concentration (PFOS and PFHxS at 167.55% and 122.62%, respectively). Again, extraction from the surface of plastics was more successful. Pesticides simazine, atrazine, and its metabolites were extracted with high efficiency (approximately 90%–120% recovery). PFAS were extracted with minimal signal enhancement compared to muscle tissue samples. Pharmaceuticals carbamazepine, atenolol, ibuprofen, and tramadol had recoveries between 100% and 160%, with the highest signal enhancement of the group seen in atenolol. Significant signal enhancement was seen across two pesticides, tebuconazole and MCPA.

The QuEChERS kit used contained sachets with 6 g magnesium sulfate and 1.5 g sodium chloride. Previous studies have found a mix of 2 g magnesium sulfate and 0.5 g sodium chloride to be more advantageous for water removal than the ratio investigated here [45]. Optimization of the salt ratio needs to be investigated further. Matrix effects likely played a significant part in the signal enhancement seen across the two QuEChERS trials, indicating that measures need to be taken to minimize such effects in the chosen extraction and analytical method.

3.3 | Matrix Effects

Matrix effects occur due to compounds within the matrix of interest eluting at the same retention time as the compounds of

TABLE 1 | Matrix effects (%) calculated for all target analytes in fish liver and muscle tissue and plastics. N/F indicates the compound was not found in the matrix when extracted using the given method.

	Method 1			Method 2			Method 3			Method 4			Method 5		
Compound	**Muscle**	**Liver**	**Plastic**	**Muscle**	**Liver**	**Plastic**	**Muscle**	**Liver**	**Plastic**	**Muscle**	**Liver**	**Plastic**	**Muscle**	**Liver**	**Plastic**
Tebuconazole	2571.2	1739.3	157.1	−116.4	−121.8	90.1	−278.6	153.0	83.8	345.6	167.8	997.8	N/F	N/F	519.6
2,4-D	504.7	272.8	93.3	−77.9	−89.7	−46.6	−79.3	−58.0	58.6	848.9	2655.8	180.4	151.9	194.9	N/F
Atrazine	−34.4	−51.7	87.2	−85.0	−93.1	9.0	−90.1	−91.9	60.3	−101.1	−98.5	−31.8	−99.5	−100.4	20.1
Atrazine-desethyl	−80.2	−100.0	−29.5	−23.6	24.3	0.9	−25.5	10.7	17.8	−98.9	−97.5	−30.7	−100.4	−99.4	4.5
Atrazine-desisopropyl	−30.4	−23.2	−45.8	N/F	N/F	10.0	1.2	48.8	33.1	−93.9	−88.2	−24.5	−94.8	−96.7	24.7
Diuron	96.9	4430.2	−100.0	−64.4	−58.4	N/F	N/F	N/F	69.5	−124.5	2249.0	N/F	194.4	2763.7	N/F
MCPA	349.2	111.7	40.3	−216.8	−52.0	−62.2	−87.5	−65.4	31.0	266.9	367.1	248.7	−120.5	−80.4	688.6
Simazine	−100.0	−100.0	−100.0	−227.1	592.9	−13.7	N/F	N/F	49.4	N/F	N/F	−36.8	N/F	N/F	−18.6
Carbamazepine	−80.9	−82.5	19.1	−74.6	−64.9	−2.9	−70.4	−57.8	55.7	−94.0	−98.0	−20.8	−98.8	−98.4	20.6
Atenolol	N/F	N/F	1132.0	N/F	N/F	273.4	N/F	N/F	1065.0	−88.5	−134.4	335.2	−97.4	−106.9	52.8
Hydrochlorothiazide	N/F	N/F	N/F	302.6	1600.9	N/F	649.8	2044.2	N/F	N/F	N/F	N/F	N/F	N/F	N/F
Diclofenac	−100.0	−100.0	−100.0	−14570.4	−342.8	N/F	N/F	965.4	39.0	−71.3	−656.9	N/F	−132.8	400.2	N/F
Ibuprofen	−100.0	−100.0	65.7	N/F	N/F	−10.4	N/F	N/F	30.2	−36.0	−71.0	−17.2	N/F	−73.8	−4.7
Tramadol	181.6	162.1	1139.7	N/F	N/F	75.6	N/F	N/F	2451.9	−90.6	−100.7	−3.0	−95.6	−96.0	39.1
PFOS	26.0	87.8	87.8	−136.0	−170.5	−4.7	32.3	217.2	65.4	630.4	1123.3	76.3	484.4	44.6	16.8
PFOA	−82.3	−88.2	−34.0	−773.8	−857.9	−12.3	−79.3	−47.1	64.6	1367.7	1658.8	28.0	610.3	98.0	−13.3
PFHxS	−17.5	−26.2	42.1	−187.1	−221.3	1.7	−18.3	8.5	55.5	715.3	573.3	61.4	262.2	8.1	7.9

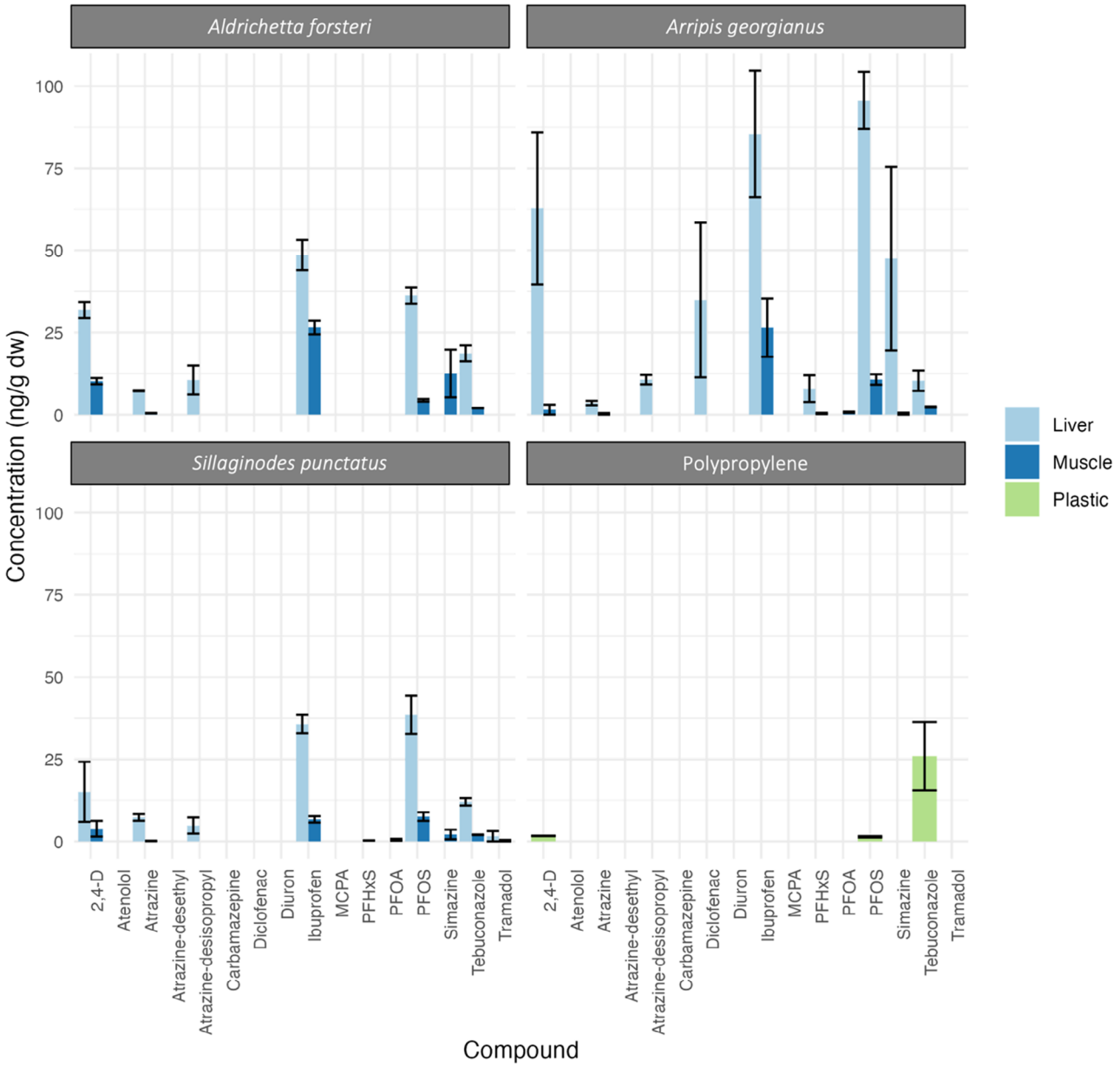


FIGURE 4 | Concentrations of target pharmaceuticals, pesticides, and PFAS in *Aldrichetta forsteri*, *Arripis georgianus*, *Sillaginodes punctatus*, and polypropylene plastic samples collected along the Adelaide metropolitan coastline (South Australia). All concentrations are expressed in ng/g dry weight.

interest, therefore, enhancing or suppressing the signal. Ideally, interfering compounds should be eliminated during sample preparation while preserving the analytes of interest [36]. A negative matrix effects value indicates signal suppression has occurred due to the presence of the matrix, whereas a positive matrix effects value indicates signal enhancement has occurred. Matrix effects varied within the different matrix types and between trials (Table 1). Matrix effects were most stable (closest to zero across compounds) in Method 3 for muscle tissue, Method 2 for liver tissue, and Method 2 for plastics. Matrix effects were higher for muscle and liver tissues than plastics. This was expected due to the complexity of the muscle and tissue extract and the coelution of lipids and proteins in the extraction process [35, 67]. Though a pre- and postspike extraction calculation was employed, the presence of signal enhancement was still evident. Due to the complex and dynamic nature of environmental samples, it can be difficult to completely account for their presence, with other potential effects leading to signal enhancement.

3.4 | Method Recommendation

The LC-MS method was not advantageous for the analysis of E1, E2, EE2, and amoxicillin and hydrochlorothiazide was not extracted successfully across any of the methods trialed; therefore, the methods were not considered further. Though the optimized method was not suitable for these compounds, the documentation of the development process and subsequent exclusions provides insight for further method development. Based on our results, Trial 2 (ultrasonic extraction with ACN and

Bond Elut SPE) was the most successful of the tested methods for the extraction of pesticides, PFAS, and pharmaceutical contaminants from fish tissues and plastics collected from marine environments. Trial 2 included the use of ultrasonic extraction with ACN as the extracting solvent, followed by a microcentrifuge step to elute lipids and proteins from the sample, then Bond Elut SPE to further concentrate analytes. For muscle samples, 5 of the 16 remaining compounds were recovered within the successful range of recovery (PFOS, PFOA, PFHxS, atrazine-desethyl, and diuron; 48.47%–82.2%) whereas for liver samples, 8 of the 16 compounds were recovered within the successful range (2,4-D, atrazine-desethyl, diuron, MCPA, diclofenac, PFOS, PFOA, and PFHxS; 50.72%–130.80%). Extraction from the plastics matrix was the most successful, with 10 of the 16 compounds within the successful range (2,4-D, atrazine, atrazine-desethyl, atrazine-desisopropyl, simazine, carbamazepine, ibuprofen, PFOS, PFOA, and PFHxS; 53.43%–111.36%). The inclusion of the microcentrifuge step along with extraction using ACN was advantageous in reducing the matrix effects, allowing for increased recoveries of the compounds of interest. This extraction method is quick and relatively simple and can be applied to many samples in a short timeframe. mLODs across the compounds were between 0.02 (tramadol in plastic) and 4.81 ppb (MCPA in fish tissue). mLOQs were low for most compounds and ranged from 0.06 (tramadol in plastic) to 14.60 ppb (MCPA in fish tissue; Table S4).

3.5 | Application to Environmental Samples

The optimized and validated method was tested on unknown samples of seafood and plastic samples from the environment. At least one of the PFAS were detected at concentrations above LOQ in fish tissues (Figure 4), which matches findings by Gaylard [68] where PFAS were found at concentrations up to c. 85 ng/g in South Australian waters. Pesticides 2,4-D, atrazine, tebuconazole, and simazine were also detected in all fish tissue samples with increased levels present in *A. georgianus* liver. Pharmaceutical ibuprofen was present in all fish tissues, again with increased levels seen in *A. georgianus*. Diclofenac was present only in *A. georgianus* liver tissues and exhibited a high degree of variability across samples (standard error of 23.5 ng/g with $n = 3$). PFHxS and PFOA were present in 53% and 100% of fish tissues, respectively, though concentrations were low (< 1 ng/g or < LOQ). Overall, liver tissues exhibited higher concentrations than muscle across all samples where concentrations were above the LOD for both tissues. In some cases, compounds were only detected in the liver. These trends of increased concentration in liver match the general trend in the literature [59, 69, 70], with the method demonstrating efficacy in analyzing environmental samples for a variety of compounds and across a range of concentrations. Only three compounds were detected in plastic, namely PFOS, 2,4-D, and tebuconazole, the latter have the highest mean concentration across all sample types investigated (Figure 4). Chemical contaminants sorb on plastics at differing rates influenced by salinity, pH, weathering of the plastic, size, and shape [71]. The developed method provides a platform to investigate different plastic types and their relative abilities to sorb chemical compounds, as well as how compounds carried by plastics may exacerbate biota contamination.

4 | Conclusion

Five common extraction methods were trialed for the extraction and quantification of 21 pesticides, PFAS, and pharmaceuticals in fish tissues and plastics from marine environments. Of these 21 compounds, 16 were efficiently quantified using LC-MS. All compounds unable to be quantified using the LC-MS method were pharmaceuticals (three estrogenics, one antibiotic, and one drug for the treatment of hypertension). The five trials had varying results across matrices and compounds of interest, with Trial 2 comprising of ultrasonic extraction with an ACN solvent, a microcentrifuge step to remove lipids and proteins, followed by SPE using Bond Elut Carbon cartridges (Agilent Technologies, USA) outperforming the other methods.

This analytical method was then successfully applied for the detection of target compounds from fish and plastics collected along the Adelaide metropolitan coastline (South Australia) and can be used more broadly to determine the presence of contaminants in marine environments across a range of matrices. As interest continues to grow in the diversity and potential impacts of emerging contaminants on marine environments, it is integral that multiresidue analytical methods are implemented. By looking at emerging contaminants in matrices such as fish tissues and plastic pollutants, using the same multiresidue approach, a better understanding of the pathways of exposure to marine species can be gained, including links between the presence of chemicals and plastics in the environment and their associated effects on biota.

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Ethics Statement

Samples were procured and processed from commercial seafood markets (scavenging) and as such did not require Animal Ethics Committee Approval in accordance with the Australian NHMRC Code for the Care and Use of Animals for Scientific Purposes. The University of Adelaide's Animal Ethics Committee was notified and no further AEC clearance required.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.